Amazon Inventory Reconciliation Using AI

MINI PROJECT

Machine Learning | 12-10-2025

Problem Statement

- Inventory reconciliation is the process of verifying whether the physical count of items matches digital inventory records. Manual reconciliation is time-consuming and prone to human error, especially in large-scale retail or warehouse systems.

- The goal of this project is to automate this reconciliation process using computer vision and machine learning — by training models to identify and count items in images of storage bins or shelves and reconcile them with expected quantities.

Approach

The solution integrates classical machine learning and deep learning to balance interpretability and accuracy:

- SVM is used with engineered visual features for lightweight, explainable classification and baseline benchmarking.
- CNN is trained end-to-end on image data to automatically learn hierarchical spatial representations for superior accuracy.
  Both models operate on the same dataset, allowing a direct performance comparison between traditional feature-based learning and deep feature extraction.

Implementation Overview

1. Dataset & Preprocessing

    - Images of items are stored in organized folders with corresponding metadata (train and validation JSON files).
    - Each image is resized to 224×224 pixels and normalized.
    - For SVM: images are flattened (224×224×3 = 150,528 features) and reduced to 5 principal components via PCA.
    - For CNN: raw images are fed directly into the model pipeline.

2. Model Design
    - SVM Pipeline:
      Input → PCA → SVM (RBF kernel, C=100, γ=1e-7)
      Trained in multiple folds with class weighting for imbalance handling.

    - CNN Pipeline
      Input → Convolutional and pooling layers → Dense layers → Softmax output.
      Implemented using TensorFlow/Keras with ReLU activation, Adam optimizer, and categorical cross-entropy loss.

3. Training & Evaluation
    - Data are divided into training and validation splits.
    - Both models generate confusion matrices, classification reports, and accuracy scores.
    - SVM serves as the baseline; CNN demonstrates the improvement achievable with deep feature learning.

Results & Observations

- SVM: Performs adequately on simple datasets but struggles with high intra-class variability.
- CNN: Shows significantly better accuracy due to automatic feature learning and spatial understanding.
- PCA-based dimensionality reduction improves SVM runtime but limits expressiveness compared to CNN's learned filters.
- Confusion matrix visualization highlights consistent misclassifications in visually similar item categories.

Conclusions

The project demonstrates that deep learning (CNN) substantially enhances the accuracy of automated inventory reconciliation over traditional SVM approaches. However, SVM remains valuable for quick inference on resource-constrained devices. Combining both—using CNN for feature extraction and SVM for final classification—offers a promising hybrid direction for future work.